

Lecture 18: Misspecification

Economics 326 — Introduction to Econometrics II

Vadim Marmer, UBC

Strong exogeneity and the CEF

- Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + U_i.$$

- When the errors are **strongly exogenous**, i.e., $E[U_i | X_i] = 0$, the linear regression model defines the **CEF** of Y conditional on X :

$$\begin{aligned} \text{CEF}_Y(X_i) &\equiv E[Y_i | X_i] \\ &= E[\beta_0 + \beta_1 X_i + U_i | X_i] \\ &= \beta_0 + \beta_1 X_i + E[U_i | X_i] \\ &= \beta_0 + \beta_1 X_i. \end{aligned}$$

Weak exogeneity

- Consider the linear regression model with $E[U_i] = 0$:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + U_i, \\ E[U_i] &= 0 \end{aligned}$$

- Suppose the errors are only **weakly exogenous**:

$$E[U_i X_i] = 0.$$

- In this case,

$$\text{CEF}_Y(X_i) \neq \beta_0 + \beta_1 X_i.$$

- Question:** What does the econometrician estimate when running a linear regression and the regressors are *not* strongly exogenous?

Misspecified CEF and the regression error

- Suppose that

$$E[Y_i | X_i] = g(X_i),$$

where g is some unknown *nonlinear* function. Thus, the **true** CEF is $g(X_i) \neq \beta_0 + \beta_1 X_i$.

- Define $V_i = Y_i - E[Y_i | X_i]$, so the true model is $Y_i = g(X_i) + V_i$ with $E[V_i | X_i] = 0$.
- Adding and subtracting $\beta_0 + \beta_1 X_i$:

$$\begin{aligned} Y_i &= g(X_i) + V_i \\ &= \beta_0 + \beta_1 X_i - \beta_0 - \beta_1 X_i + g(X_i) + V_i \end{aligned}$$

- Rearranging, $Y_i = \beta_0 + \beta_1 X_i + U_i$, where

$$U_i = V_i + g(X_i) - \beta_0 - \beta_1 X_i.$$

- Can we find β_0 and β_1 so that $E[U_i] = 0$ and $E[X_i U_i] = 0$? If yes, how can we interpret such β_0 and β_1 ?

Conditions for weak exogeneity

$$U_i = V_i + g(X_i) - \beta_0 - \beta_1 X_i.$$

- Evaluating $E[U_i]$:

$$\begin{aligned} E[U_i] &= E[V_i + g(X_i) - \beta_0 - \beta_1 X_i] \\ &= E[V_i] + E[g(X_i) - \beta_0 - \beta_1 X_i] \\ &= E[g(X_i) - \beta_0 - \beta_1 X_i], \end{aligned}$$

- For the covariance condition:

$$\begin{aligned} E[U_i X_i] &= E[(V_i + g(X_i) - \beta_0 - \beta_1 X_i) X_i] \\ &= E[V_i X_i] + E[(g(X_i) - \beta_0 - \beta_1 X_i) X_i] \\ &= E[(g(X_i) - \beta_0 - \beta_1 X_i) X_i]. \end{aligned}$$

- Thus, to have $E[U_i] = E[U_i X_i] = 0$, we need to find β_0 and β_1 such that

$$\begin{aligned} E[g(X_i) - \beta_0 - \beta_1 X_i] &= 0, \\ E[(g(X_i) - \beta_0 - \beta_1 X_i) X_i] &= 0. \end{aligned}$$

Linear approximation of the CEF

- Consider the following approximation problem:

$$\min_{b_0, b_1} E[(g(X_i) - b_0 - b_1 X_i)^2].$$

- We are approximating the CEF with a linear function.
- Among the linear functions, we are looking for the **best** linear approximation in the **mean squared error (MSE)** sense.

Regression as best linear approximation

- Let $(\beta_0, \beta_1) = \arg \min_{b_0, b_1} \text{MSE}(b_0, b_1)$, where $\text{MSE}(b_0, b_1) = E[(g(X_i) - b_0 - b_1 X_i)^2]$.
- The first-order conditions are:

$$\begin{aligned} \frac{\partial \text{MSE}}{\partial b_0} &= -2E[g(X_i) - \beta_0 - \beta_1 X_i] = 0, \\ \frac{\partial \text{MSE}}{\partial b_1} &= -2E[(g(X_i) - \beta_0 - \beta_1 X_i) X_i] = 0. \end{aligned}$$

- These are the conditions for weak exogeneity derived earlier, so

$$E[U_i] = 0 \text{ and } E[U_i X_i] = 0.$$

- Thus, the linear regression model gives us the best linear approximation of the CEF (in the MSE sense).

Misspecification and heteroskedasticity

- We have

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 X_i + U_i, \\U_i &= V_i + g(X_i) - \beta_0 - \beta_1 X_i.\end{aligned}$$

- Suppose that the “true” error V_i is homoskedastic: $E[V_i^2 | X_i] = \sigma_V^2$ for all X_i .
- U_i is **heteroskedastic** if $g(X_i) \neq \beta_0 + \beta_1 X_i$:

$$\begin{aligned}E[U_i^2 | X_i] &= E[(V_i + g(X_i) - \beta_0 - \beta_1 X_i)^2 | X_i] \\&= E[V_i^2 + (g(X_i) - \beta_0 - \beta_1 X_i)^2 | X_i] \\&\quad + E[2V_i(g(X_i) - \beta_0 - \beta_1 X_i) | X_i] \\&= \sigma_V^2 + (g(X_i) - \beta_0 - \beta_1 X_i)^2.\end{aligned}$$